FastAPI Learn Tutorial - User Guide

First Steps

The simplest FastAPI file could look like this:

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI()


@app.get("/")
async def root():
    return {"message": "Hello World"}
```

Copy that to a file `main.py`.

Run the live server:

```bash
fast →
$ fastapi dev

  FastAPI   Starting development server 🚀

            Searching for package file structure from directories
            with __init__.py files
            Importing from /home/user/code/awesomeapp

   module   🐍 main.py

     code   Importing the FastAPI app object from the module with
            the following code:

            from main import app

      app   Using import string: main:app

   server   Server started at http://127.0.0.1:8000
   server   Documentation at http://127.0.0.1:8000/docs

      tip   Running in development mode, for production use:
            fastapi run

            Logs:

     INFO   Will watch for changes in these directories:
            ['/home/user/code/awesomeapp']
     INFO   Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C
            to quit)
     INFO   Started reloader process [383138] using WatchFiles
     INFO   Started server process [383153]
     INFO   Waiting for application startup.
```

```
INFO     Application startup complete.
```

In the output, there's a line with something like:

```
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

That line shows the URL where your app is being served on your local machine.

Check it

Open your browser at http://127.0.0.1:8000.

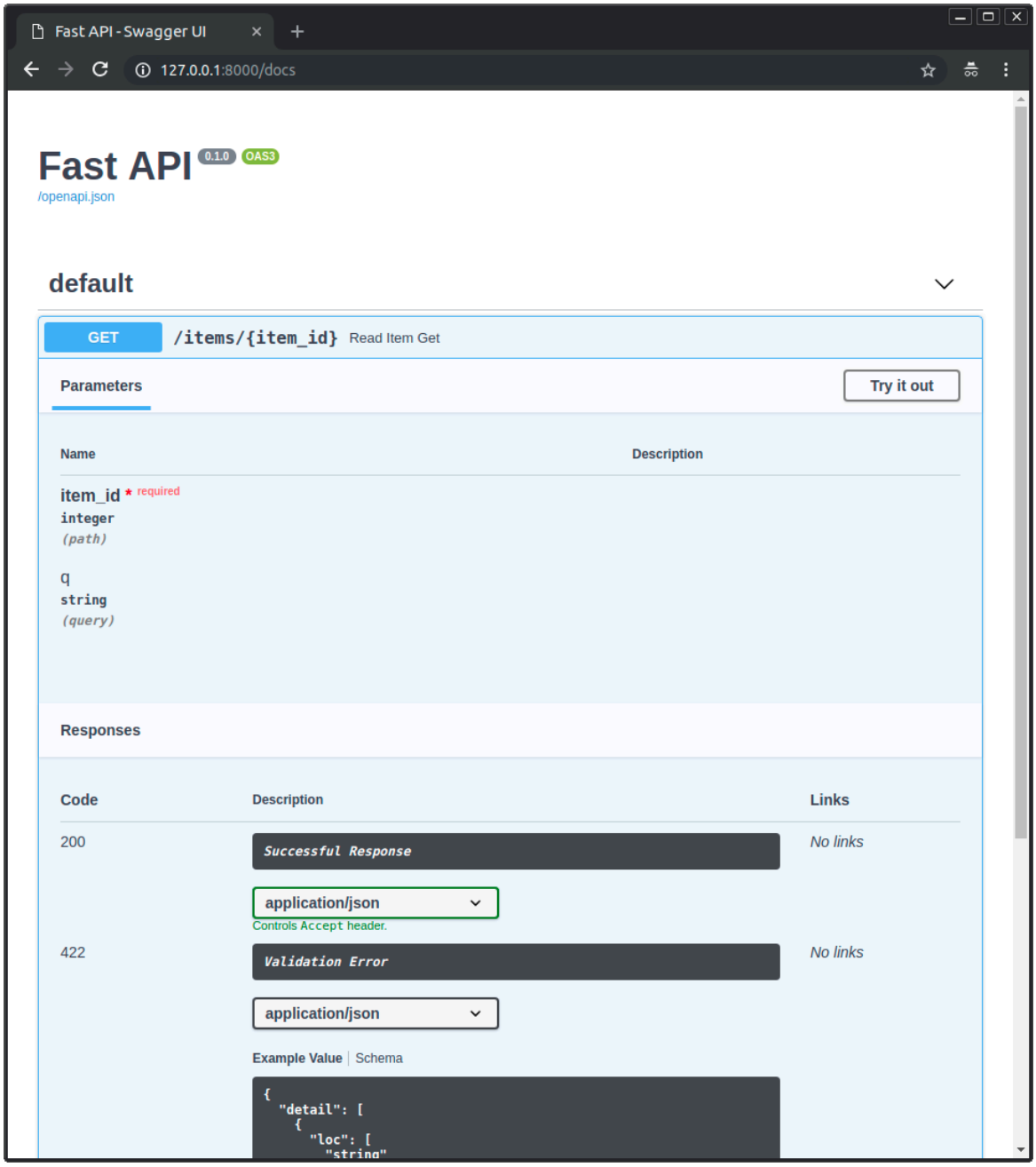
You will see the JSON response as:

```
{"message": "Hello World"}
```

Interactive API docs

Now go to http://127.0.0.1:8000/docs.

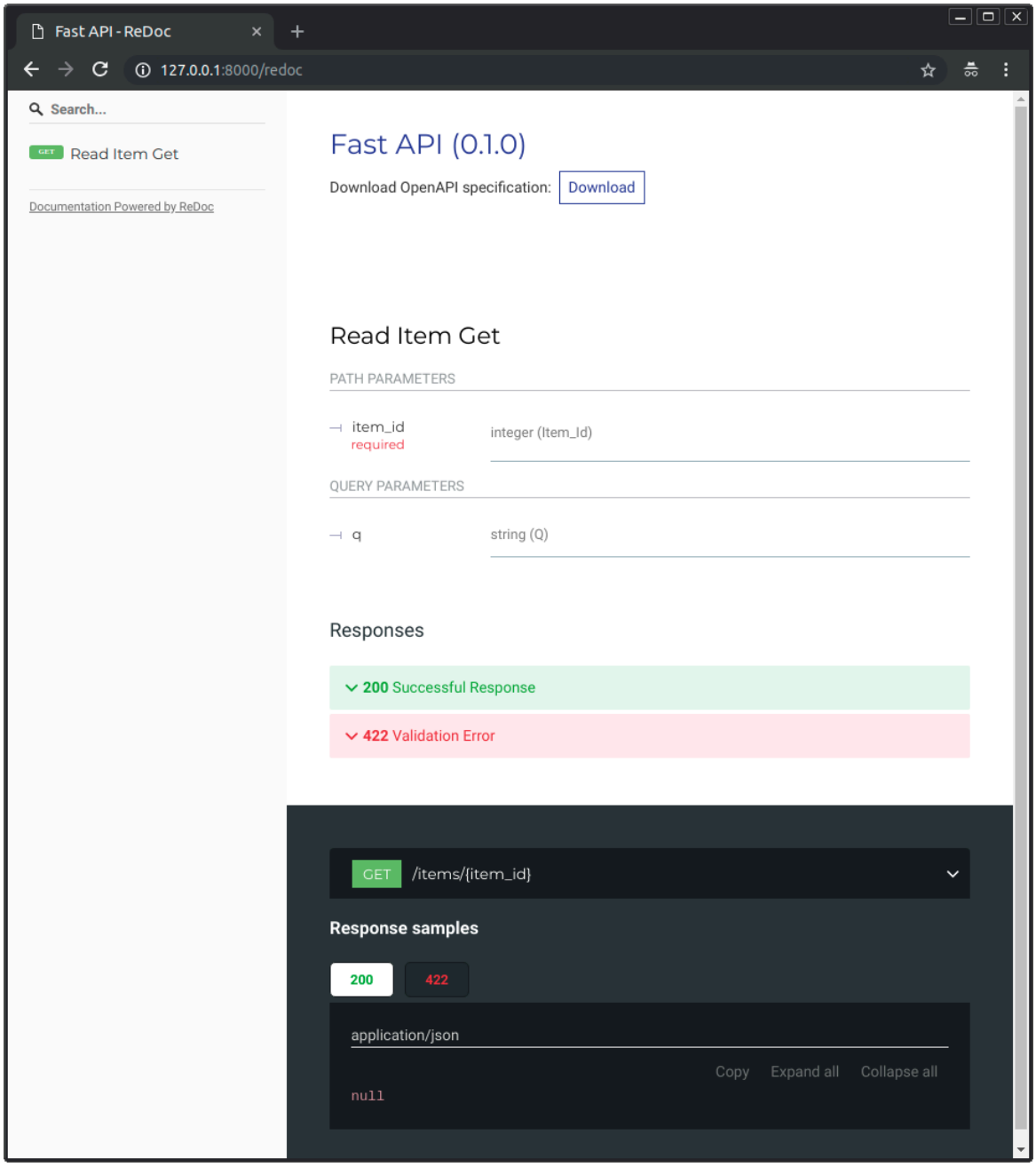
You will see the automatic interactive API documentation (provided by Swagger UI):



Alternative API docs

And now, go to http:// 1 2 7 . 0 . 0 . 1 : 8 0 0 0 /redoc.

You will see the alternative automatic documentation (provided by ReDoc):



OpenAPI

FastAPI generates a "schema" with all your API using the **OpenAPI** standard for defining APIs.

"Schema"

A "schema" is a definition or description of something. Not the code that implements it, but just an abstract description.

API "schema"

In this case, OpenAPI is a specification that dictates how to define a schema of your API.

This schema definition includes your API paths, the possible parameters they take, etc.

Data "schema"

The term "schema" might also refer to the shape of some data, like a JSON content.

In that case, it would mean the JSON attributes, and data types they have, etc.

OpenAPI and JSON Schema

OpenAPI defines an API schema for your API. And that schema includes definitions (or "schemas") of the data sent and received by your API using **JSON Schema**, the standard for JSON data schemas.

Check the `openapi.json`

If you are curious about how the raw OpenAPI schema looks like, FastAPI automatically generates a JSON (schema) with the descriptions of all your API.

You can see it directly at: http://127.0.0.1:8000/openapi.json.

It will show a JSON starting with something like:

```
{
    "openapi": "3.1.0",
    "info": {
        "title": "FastAPI",
        "version": "0.1.0"
    },
    "paths": {
        "/items/": {
            "get": {
                "responses": {
                    "200": {
                        "description": "Successful Response",
                        "content": {
                            "application/json": {



...
```

What is OpenAPI for

The OpenAPI schema is what powers the two interactive documentation systems included.

And there are dozens of alternatives, all based on OpenAPI. You could easily add any of those alternatives to your application built with **FastAPI**.

You could also use it to generate code automatically, for clients that communicate with your API. For example, frontend, mobile or IoT applications.

Configure the app `entrypoint` in `pyproject.toml`

You can configure where your app is located in a `pyproject.toml` file like:

```
[tool.fastapi]
entrypoint = "main:app"
```

That `entrypoint` will tell the `fastapi` command that it should import the app like:

```
from main import app
```

If your code was structured like:

```
.
├── backend
│   ├── main.py
│   ├── __init__.py
```

Then you would set the `entrypoint` as:

```
[tool.fastapi]
entrypoint = "backend.main:app"
```

which would be equivalent to:

```
from backend.main import app
```

`fastapi dev` with path

You can also pass the file path to the `fastapi dev` command, and it will guess the FastAPI app object to use:

```
$ fastapi dev main.py
```

But you would have to remember to pass the correct path every time you call the `fastapi` command.

Additionally, other tools might not be able to find it, for example the VS Code Extension or FastAPI Cloud, so it is recommended to use the `entrypoint` in `pyproject.toml`.

Deploy your app (optional)

You can optionally deploy your FastAPI app to FastAPI Cloud, go and join the waiting list if you haven't. 🚀

If you already have a **FastAPI Cloud** account (we invited you from the waiting list 😉), you can deploy your application with one command.

Before deploying, make sure you are logged in:

```bash
```

Then deploy your app:

```bash
```

That's it! Now you can access your app at that URL. ✨

Recap, step by step

Step 1 : import `FastAPI`

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI()


@app.get("/")
async def root():
    return {"message": "Hello World"}
```

`FastAPI` is a Python class that provides all the functionality for your API.

> **Technical Details**
>
> `FastAPI` is a class that inherits directly from `Starlette`.
>
> You can use all the Starlette functionality with `FastAPI` too.

Step 2 : create a `FastAPI` "instance"

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI()


@app.get("/")
async def root():
    return {"message": "Hello World"}
```

Here the `app` variable will be an "instance" of the class `FastAPI`.

This will be the main point of interaction to create all your API.

Step 3 : create a *path operation*

Path

"Path" here refers to the last part of the URL starting from the first `/`.

So, in a URL like:

```
https://example.com/items/foo
```

...the path would be:

```
/items/foo
```

> **Info**
>
> A "path" is also commonly called an "endpoint" or a "route".

While building an API, the "path" is the main way to separate "concerns" and "resources".

Operation

"Operation" here refers to one of the HTTP "methods".

One of:

- `POST`
- `GET`
- `PUT`
- `DELETE`

...and the more exotic ones:

- `OPTIONS`
- `HEAD`
- `PATCH`
- `TRACE`

In the HTTP protocol, you can communicate to each path using one (or more) of these "methods".

When building APIs, you normally use these specific HTTP methods to perform a specific action.

Normally you use:

- `POST`: to create data.
- `GET`: to read data.
- `PUT`: to update data.
- `DELETE`: to delete data.

So, in OpenAPI, each of the HTTP methods is called an "operation".

We are going to call them "**operations**" too.

Define a *path operation decorator*

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI()


@app.get("/")
async def root():
    return {"message": "Hello World"}
```

The `@app.get("/")` tells **FastAPI** that the function right below is in charge of handling requests that go to:

- the path `/`
- using a `get` *operation*

> **`@decorator` Info**
>
> That `@something` syntax in Python is called a "decorator".
>
> You put it on top of a function. Like a pretty decorative hat (I guess that's where the term came from).
>
> A "decorator" takes the function below and does something with it.
>
> In our case, this decorator tells **FastAPI** that the function below corresponds to the **path** `/` with an **operation** `get`.
>
> It is the "**path operation decorator**".

You can also use the other operations:

- `@app.post()`
- `@app.put()`
- `@app.delete()`

And the more exotic ones:

- `@app.options()`
- `@app.head()`
- `@app.patch()`
- `@app.trace()`

> **Tip**
>
> You are free to use each operation (HTTP method) as you wish.
>
> **FastAPI** doesn't enforce any specific meaning.
>
> The information here is presented as a guideline, not a requirement.
>
> For example, when using GraphQL you normally perform all the actions using only `POST` operations.

Step 4 : define the **path operation function**

This is our "**path operation function**":

- **path**: is `/`.
- **operation**: is `get`.
- **function**: is the function below the "decorator" (below `@app.get("/")`).

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI()


@app.get("/")
async def root():
    return {"message": "Hello World"}
```

This is a Python function.

It will be called by **FastAPI** whenever it receives a request to the URL "`/`" using a `GET` operation.

In this case, it is an `async` function.

You could also define it as a normal function instead of `async def`:

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI()


@app.get("/")
def root():
    return {"message": "Hello World"}
```

Note

If you don't know the difference, check the Async: *"In a hurry?"*.

Step 5: return the content

Python 3.10+

```python
from fastapi import FastAPI

app = FastAPI()


@app.get("/")
async def root():
    return {"message": "Hello World"}
```

You can return a `dict`, `list`, singular values as `str`, `int`, etc.

You can also return Pydantic models (you'll see more about that later).

There are many other objects and models that will be automatically converted to JSON (including ORMs, etc). Try using your favorite ones, it's highly probable that they are already supported.

Step 6: Deploy it

Deploy your app to **FastAPI Cloud** with one command: `fastapi deploy`. 🎉

About FastAPI Cloud

FastAPI Cloud is built by the same author and team behind **FastAPI**.

It streamlines the process of **building**, **deploying**, and **accessing** an API with minimal effort.

It brings the same **developer experience** of building apps with FastAPI to **deploying** them to the cloud. 🎉

FastAPI Cloud is the primary sponsor and funding provider for the *FastAPI and friends* open source projects. ✨

Deploy to other cloud providers

FastAPI is open source and based on standards. You can deploy FastAPI apps to any cloud provider you choose.

Follow your cloud provider's guides to deploy FastAPI apps with them. 🤓

Recap

- Import `FastAPI`.
- Create an `app` instance.
- Write a **path operation decorator** using decorators like `@app.get("/")`.
- Define a **path operation function**; for example, `def root(): ...`.
- Run the development server using the command `fastapi dev`.
- Optionally deploy your app with `fastapi deploy`.